

Erdős Problem #340

EP-340	L1 - Tractable	Ouvert	Number Theory
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Énoncé original (anglais, reproduction mot pour mot)

Let $A=\{1,2,4,8,13,21,31,45,66,81,97,\ldots\}$ be the greedy Sidon sequence: we begin with 1 and iteratively include the next smallest integer that preserves the Sidon property (i.e. there are no non-trivial solutions to $a+b=c+d$). What is the order of growth of A ? Is it true that $\left|A \cap \{1,\ldots,N\}\right| \gg N^{1/2-\epsilon}$ for all $\epsilon>0$ and large N ?

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